

CSA05-DATABASE MANAGEMENT SYSTEMS

LAB OBSERVATION

EXPERIMENT:16

AIM:

To implement function using program in MySQL.

Procedure and Syntax

Functions:

A function is a subprogram that computes a value.

Creating a function

The CREATE FUNCTION statement is also used in MySQL to support UDFs (user-defined functions). A UDF can be regarded as an external stored function.

MySQL stored function syntax

```
CREATE FUNCTION function_name(param1,param2,...)
```

```
RETURNS datatype
```

```
[NOT] DETERMINISTIC
```

```
.....
```

```
statements
```

```
....
```

Problem 1:

Create a function that returns the level of a customer based on credit limit.

(Use the IF statement to determine the credit limit).

If credit limit > 50000 then customer_level = PLATINUM

If credit limit >= 10000 AND credit limit <= 50000 then customer_level = GOLD

If credit limit credit limit < 10000 then customer_level = SILVER

RECURSION in Mysql Procedures

Mysql version should be >= 5.

Have to set system parameters. This means putting the recursion count limit.

```
SET @@GLOBAL.max_sp_recursion_depth = 255;
```

```
SET @@session.max_sp_recursion_depth = 255;
```

Problem 2

Write a recursive MySQL procedure compute the factorial of a number .

SQL QUERIES:

```

-> END \\\
-> END TL\
-> SET b'список_файлов' = ,217АЕВ,;
-> IFGE
-> SET b'список_файлов' = ,00ГД,;
-> IFGE11 A'соединити' = 10000 ИМЕИ
-> SET b'список_файлов' = ,БГВ11ИИИ,;
-> IF A'соединити' > 20000 ИМЕИ
-> ЕНОВ список_ИМЕНЕ список_ИМЕНЕ = b'список_ИМЕНЕ;
-> SETFC1 соединити ИИО A'соединити
-> DECTIME A'соединити, DECTIME(10*5);
-> BEGIN
-> ON1 b'список_файлов' АВВЧВВ(50)
-> ИИ b'список_ИМЕНЕ ИИИ
\клетр> CHEVJE BPOCEBME COTCписок_файлов(
\клетр> DECTIME \\\

2 LAMP TU 201 (0'00 sec)
+-----+
| 2 | ЕВВ | 55 |
| 4 | Д1А10 | 50 |
| 3 | С1010 | 10 |
| 5 | Б111 | 51 |
| 1 | В111 | 50 |
+-----+
| 10 | ИИИ | 100 |
+-----+
\клетр> СУГГ COTCсоединити():
\клетр> DECTIME :
\клетр> ОК* 0 LAMP 2100000 (0'01 sec)
-> END \\\
-> SETFC1 * ЕНОВ соединити:
-> BEGIN
\клетр> CHEVJE BPOCEBME COTCсоединити()
\клетр> DECTIME \\\

```

RESULT:

Thus the Functions using program in MySQL executed successfully .